

Analyzing Topological Solitons in the Kitaev Spin Chain using Density Matrix Renormalization Group

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A thesis presented for the degree of
Bachelor of Science



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October 21, 2025

Abstract

The one-dimensional Kitaev spin chain is a fundamental model for studying topological phases in condensed matter physics. An ongoing area of research has been trying to understand the emergence and behavior of topological solitons in this system, particularly how their formation is influenced by external parameters like magnetic fields. Prior research has numerically observed soliton phases in the Kitaev chain under specific field conditions [1]. Building on this foundation, the present thesis aims to independently verify these findings for the spin- $\frac{1}{2}$ Kitaev chain subjected to an external magnetic field at an angle using the finite Density Matrix Renormalization Group (DMRG) method. Topological solitons, or stable nonlinear localized wave packets, are then identified from the calculated spin expectation profiles. Our numerical results confirm the existence of a stable yet unexpected topological soliton phase. This phase is predominantly observed when the magnetic field is aligned near the xy -plane and for a specific range of field strengths h and in-plane angles ϕ_{xy} . The calculated phase boundaries are consistent with the previously reported predictions. This thesis thus validates the presence of the solitonic phase and demonstrates the effectiveness of the DMRG approach for their characterization in the Kitaev model.

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Chapter 1

Introductory Material

1.1 Introduction

The study of one-dimensional quantum spin systems has, for nearly a century, been at the center of theoretical condensed matter physics. Spin chains, although structurally simple, exhibit a plethora of phenomena that elucidate the behavior of quantum many-body systems. As such, spin chains have been used both as idealized models for specific condensed matter systems and as theoretical testing grounds for the development of new tools, both analytical and computational. Despite the simplistic structure of spin chains, many systems do not yet have exact solutions, which provides a rich area of study for mathematicians and physicists alike.

A landmark in the theoretical development of spin chains was the *Bethe Ansatz* for the Heisenberg model, pioneered by Hans Bethe in 1931 [2]. The Bethe Ansatz is a method for exact diagonalization of the Hamiltonian for a one-dimensional chain of spin- $\frac{1}{2}$ particles with isotropic nearest neighbor interactions. The Bethe Ansatz also inadvertently introduced the notion of quantum integrability, an area of mathematics that is still being studied today [3]. In subsequent decades, generalizations of the Heisenberg model, including the anisotropic XXZ and XYZ chains, were explored to study critical behavior, spectral gaps, and symmetry

breaking [4, 5, 6, 7].

Spin chains have also provided a natural setting for the study of quantum phase transitions. For instance, the transverse-field Ising chain has been the go-to model for studying systems with quantum phase transitions [8]. The transverse-field Ising chain is also deeply connected with conformal field theories (CFTs) [9], a fact that has been exploited to explain entanglement scaling laws and low-energy effective theories of spin chains [10, 11].

From the 1980s onwards, interest in higher-spin chains (e.g. spin-1 or spin-3/2) broadened the work being done in theoretical condensed matter physics. In particular, the Haldane conjecture [12] suggested that the integer-spin Heisenberg antiferromagnetic (AFM) spin chain exhibited gapped excitation spectra versus the ungapped fractional integer-spin Heisenberg chain. The Haldane conjecture was revolutionary in understanding topological phases in quantum spin systems, and as a result, Duncan Haldane was awarded the 2016 Nobel Prize in Physics. Following Haldane’s work, integer-spin chains such as the Affleck-Kennedy-Lieb-Tasaki (AKLT) model were shown to have symmetry-protected topological (SPT) phases [13, 14].

The 21st century has seen renewed interest in spin chains, both in the theoretical and experimental setting. In the theoretical setting, spin chains have continued to play a central role in the study of integrability [15, 16], the AdS/CFT correspondence [17], and even supersymmetric string theory [18, 19]. On the experimental side spin chains can be used to model linear excitations called *magnons* in the material CrSBr [personal correspondence with Axel Saenz Rodriguez]. Recent work [20] has substantiated this perspective that spin chains can model the properties (such as magnon dynamics) of CrSBr or other 1D or quasi-2D materials.

Despite the importance of integrable models, most physically relevant spin chains are non-integrable— they do not admit exact solutions. It is in these cases that analytical methods either require significant approximations to be made or do not exist at all. The exponential growth of the Hilbert space with system size ($\dim(\mathcal{H}) \sim 2^L$ for spin- $\frac{1}{2}$ chains) makes exact diagonalization intractable for long chains in general. The limitations of these non-integrable

spin chains along with their striking physical importance has spurred the development of numerical techniques specifically tailored to low-dimensional quantum systems.

One such technique is the Density Matrix Renormalization Group (DMRG), introduced by Steven White in 1992 [21]. The method was initially developed as an improvement to Wilson's numerical renormalization group [22], where the key improvement was the utilization of the reduced density matrix to determine the optimal truncation of the Hilbert space. DMRG achieves remarkable accuracy for ground states of one-dimensional gapped systems by targeting low-entanglement subspaces. It has since been reinterpreted in the language of tensor networks, specifically matrix product states (MPS), which elucidate its variational structure and allow generalizations to time evolution (so-called tDMRG), finite temperature, and higher dimensions [23, 24, 25].

The present work applies the DMRG algorithm to study the ground state properties and an emergent soliton phase in the Kitaev spin chain. We classify the solitons with tools from homotopy theory.

The structure of the thesis is as follows. Chapter 1 provides the necessary background for the results produced in this thesis. Sections 1.2 and 1.3 introduce spin chains and their basic statistical mechanical properties. The final section of the introductory material is Section 1.4, which introduces solitons as solutions to nonlinear PDEs, called integrable solitons, as well as basic homotopy theory and topological solitons. Chapter 2 introduces DMRG in Section 2.1 and the language in which modern DMRG is framed, Matrix Product States (MPS), in Section 2.2. The mathematical machinery for which can be found in the Appendix. We confirm the existence of a soliton phase in the spin- $\frac{1}{2}$ Kitaev chain in Section 3, and give concluding thoughts and possible future work in Section 4.

1.2 Spin Chains

The spin chains that we will study here are typically described as nearest neighbor spin-spin interaction models. In the physics literature, spin chains are used to study critical points, entanglement properties, and phase transitions of magnetic statistical mechanical systems.

In the forthcoming discussion we consider the periodic lattice with $L \in \mathbb{Z}_{>0}$ sites. Each site of the lattice is labeled by a *spin-up*, $|\uparrow\rangle$, or *spin-down*, $|\downarrow\rangle$. The labels of each site are updated as the system evolves due to local pairwise (nearest neighbor) interactions. We consider the evolution of N up-spins (particles) on the periodic lattice of length L . A *configuration* of the lattice is given as the location of the N particles $X = (x_1, x_2, \dots, x_N)$ so that $x_i \in \{0, 1, \dots, L-1\} = \Lambda$ and $x_i < x_{i+1}$. The *space of configurations*, $\mathcal{X}$, is given as

$$\mathcal{X} = \{(x_1, x_2, \dots, x_N) \mid x_i \in \Lambda, i = 1, 2, \dots, N\}. \quad (1.1)$$

The propagation of a particle on the lattice is governed by the time-dependent Schrödinger equation,

$$i\hbar \frac{d}{dt} |\Psi(t)\rangle = \mathcal{H} |\Psi(t)\rangle, \quad (1.2)$$

with $\mathcal{H}$ dependent on the chosen model. Note that the set of configurations $\mathcal{X}$ is an orthonormal basis set. For each configuration $X \in \mathcal{X}$, we take a basis vector $|X\rangle$ and construct the vector space generated by these vectors,

$$\mathbb{H} := \text{Span}\{|X\rangle \mid X \in \mathcal{X}\}. \quad (1.3)$$

We say a vector $|\Psi\rangle \in \mathbb{H}$ is a *state* if it is normalized ($\langle\Psi|\Psi\rangle = 1$). Each site $i \in \Lambda$ is associated with a two dimensional (complex) Hilbert space $\mathbb{V} = \text{Span}\{|\uparrow\rangle, |\downarrow\rangle\} \cong \mathbb{C}^2$ with basis given by up-spin and down-spin labels. Consider now the L -fold tensor product of $\mathbb{V}$, $\mathbb{V}_L = \mathbb{V}^{\otimes L}$. The basis for $\mathbb{V}_L$ is given by taking L tensors of basis elements from $\mathbb{V}$. For

example, the basis of $\mathbb{V}_2$ is given as

$$|\uparrow\rangle \otimes |\uparrow\rangle = |\uparrow\uparrow\rangle, |\uparrow\rangle \otimes |\downarrow\rangle = |\uparrow\downarrow\rangle, |\downarrow\rangle \otimes |\uparrow\rangle = |\downarrow\uparrow\rangle, |\downarrow\rangle \otimes |\downarrow\rangle = |\downarrow\downarrow\rangle. \quad (1.4)$$

The Hilbert space defined in (1.3) is a subspace of $\mathbb{V}_L$. That is, each basis element $|X\rangle \in \mathbb{H}$ is identified with a basis element of $\mathbb{V}_L$ with the position of the up-spins given as $X = (x_1, \dots, x_N) \in \mathcal{X}$. For example, we have the identification

$$|\uparrow\downarrow\rangle = |1\rangle, |\downarrow\uparrow\rangle = |2\rangle \quad (1.5)$$

for $N = 1$ and $L = 2$.

The evolution of particles in the system is dictated by *local* Hamiltonian operators. The construction of the local Hamiltonian operators will be heavily dependent on the model, but we shall try to give a general idea. We first need to define the following operators on $\mathbb{V}_L$:

$$\mathbf{S}_i^\alpha \mathbf{S}_j^\alpha = \frac{1}{2} (\mathbb{1} \otimes \dots \otimes \mathbb{1} \otimes \sigma^\alpha \otimes \mathbb{1} \otimes \dots \otimes \mathbb{1} \otimes \sigma^\alpha \otimes \mathbb{1} \otimes \dots \otimes \mathbb{1}), \quad (1.6)$$

where $\mathbb{1}$ is the identity operator on $\mathbb{V}$ and each σ^α (for $\alpha = x, y, z$) is a *Pauli matrix*, given as

$$\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (1.7)$$

It is typical that the local Hamiltonian operators are given by tensor products of the Pauli matrices defined above. A general construction may be

$$\mathcal{H} = \mathcal{J} \sum_{\langle i,j \rangle} h_{i,j}(g), \quad (1.8)$$

where the local Hamiltonian operator $h_{i,j}(g)$ is dependent on some (typically continuous) parameter $g \in \mathbb{R}$. The parameter $\mathcal{J}$ is the *coupling parameter* between sites on the lattice,

and we use the convention that $\mathcal{J} < 0$ corresponds to an antiferromagnetic spin chain, and $\mathcal{J} > 0$ corresponds to a ferromagnetic chain. It is important to note that $h_{i,j}(g)$ is an automorphism on $\mathbb{H}$, so we can restrict $h_{i,j}(g)$ to $\mathbb{H}$

1.2.1 Ising model

The 1D Ising model is the simplest spin chain model that we can study. The Hamiltonian for the Ising model is

$$\mathcal{H}_{\text{ising}} = \mathcal{J} \sum_{\langle i,j \rangle} S_i^z S_j^z + h \sum_i S_i^x, \quad (1.9)$$

where h is the strength of the magnetic field, and $\mathcal{J}$ is the usual coupling parameter, which we take to be $\mathcal{J} > 0$. Note that in the Ising model, we only consider interactions between each site in the z direction, and apply an external magnetic field only in the x direction. The Ising model does not exhibit a classical phase transition in 1D.

1.2.2 Kitaev spin chain

The Kitaev spin chain is characterized by alternating S^x and S^y terms on a honeycomb lattice [26]. The Hamiltonian for the Kitaev spin chain on a 1D lattice is given as

$$\mathcal{H} = \mathcal{J} \sum_i S_{2i+1}^x S_{2i+2}^x + S_{2i+2}^y S_{2i+3}^y - \sum_i \mathbf{h} \cdot \mathbf{S}_i, \quad (1.10)$$

where $\mathbf{h} = (\cos \phi_{xy} \cos \theta_z, \sin \phi_{xy} \cos \theta_z, \sin \theta_z)$ and $\mathbf{S}_i = (S_i^x, S_i^y, S_i^z)$. In the present thesis we consider only the spin- $\frac{1}{2}$ Kitaev chain.

1.3 The physics of spin chains

The Hilbert space (and algebraic) structures defining quantum spin chains, as discussed in Section 1.2, provide the ability to investigate the physical characteristics of spin chains. This section extends the formalism of Section 1.2 to present excitations, collective phenomena,

and the general statistical mechanics of spin chains. Our discussion will provide an overview of linear excitations formulated as quasiparticles, quantum phase transitions, the properties of critical points for making such transitions, and correlation functions.

1.3.1 Spin waves and magnons

In a quantum spin system, the ground state $|\Psi_0\rangle$ of a given Hamiltonian $\mathcal{H}$ is the eigenvector corresponding to the lowest eigenvalue E_0 . Deviations from the ground state configuration caused by low energy (linear) excitations propagate through the lattice as a collective mode called a *spin wave*. When these modes are quantized, they become quasiparticles called *magnons*.

Consider a lattice $\Lambda = \{1, 2, \dots, L\}$ of sites, each associated with a spin operator $\mathbf{S}_j = (S_j^x, S_j^y, S_j^z)$. Suppose the ground state of the system $|\Psi_0\rangle$ has some magnetic order, e.g., a ferromagnetic state $|\text{FM}\rangle = \bigotimes_{j=1}^L |S, S\rangle_j$, where $|S, m_S\rangle_j$ denotes an eigenstate of S_j^z with an eigenvalue of m_S , and here $m_S = S$. In the ferromagnetic case, we may consider deviations in an individual spin and investigate the resulting propagation. In fact, a convenient picture to have is that the propagation of a spin wave is the collective precession of individual spins, as seen in Figure 1.1. To describe the propagation in the ground state formally, we now

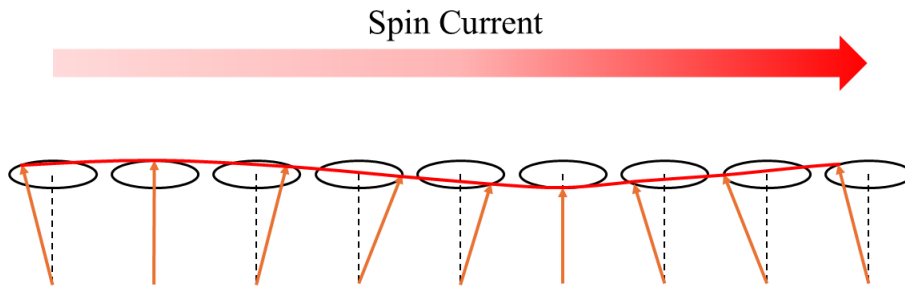


Figure 1.1: Collective precession of individual spin-moments creates the effect of a spin wave, and hence a spin current.

introduce an important transformation in condensed matter physics.

The *Holstein-Primakoff transformation* maps the spin operators at each site j (which acts on a $(2S + 1)$ -dimensional local Hilbert space) to the bosonic creation ($a_j^\dagger$) and annihilation

(a_j) operators. More precisely, the Holstein-Primakoff transformation is a map

$$\sigma : \mathcal{V}_j \rightarrow \mathcal{V}_j^*, \quad (1.11)$$

where $\mathcal{V}_j$ is the $2S + 1$ -dimensional local Hilbert space at site $j \in \Lambda$ and $\mathcal{V}_j^*$ is the infinite dimensional Fock space space at the same $j \in \Lambda$. The action of σ is given as

$$\begin{aligned} \sigma(S_j^z) &\mapsto S - a_j^\dagger a_j \\ \sigma(S_j^x + iS_j^y) &\mapsto \sqrt{2S - a_j^\dagger a_j} a_j \\ \sigma(S_j^x - iS_j^y) &\mapsto a_j^\dagger \sqrt{2S - a_j^\dagger a_j}, \end{aligned} \quad (1.12)$$

where a_j and $a_j^\dagger$ are such that $[a_j, a_k^\dagger] = \delta_{jk} \mathbb{1}$ and $[a_j, a_k] = [a_j^\dagger, a_k^\dagger] = 0$ for all j, k . The *number operator* $N_j = a_j^\dagger a_j$ counts the number of spin deviations at site j relative to the state $S_j^z = S$. The original $(2S + 1)$ spin states at a site are recovered by restricting N_j to eigenvalues $n_j \in \{0, 1, \dots, 2S\}$. Physically, the condition that $n_j \in \{0, 1, \dots, 2S\}$ means we restrict the number of bosons on each site to $0 \leq n_j \leq 2S$. Indeed, for a spin- $\frac{1}{2}$ chain the number of bosons per site is 2, corresponding to $\uparrow$ and $\downarrow$. For systems where the density of excitations is low ($\langle N_j \rangle \ll 2S$), the nonlinear terms of (1.12) can be approximated by expanding them in powers of N_j/S . If we take the leading-order approximation, which is referred to as linear spin wave theory (LSWT) in the literature [27], we have

$$\begin{aligned} \sigma(S_j^x + iS_j^y) &\approx \sqrt{2S} a_j, \\ \sigma(S_j^x - iS_j^y) &\approx \sqrt{2S} a_j^\dagger. \end{aligned} \quad (1.13)$$

Plugging the LSWT terms (1.13) to a spin Hamiltonian (like the Kitaev Hamiltonian 1.10) results in a Hamiltonian that is quadratic in the bosonic operators [28]. Diagonalizing such a Hamiltonian can be done by Fourier transforming it to momentum space. The bosonic

operators a_j and $a_j^\dagger$ under this Fourier transform become

$$\begin{aligned} a_j &= \frac{1}{\sqrt{L}} \sum_k e^{ikj} a_k, \\ a_j^\dagger &= \frac{1}{\sqrt{L}} \sum_k e^{-ikj} a_k^\dagger, \end{aligned} \tag{1.14}$$

where L is the length of the chain and k is the 1D wavevector. We implicitly assume the chain has unit lattice spacing. The Hamiltonian in momentum space thus takes the form

$$\mathcal{H} \approx E_0 + \sum_k \omega(k) a_k^\dagger a_k, \tag{1.15}$$

where E_0 is the ground state energy and $\omega(k)$ is the energy of a magnon with wavevector k [29]. What has been presented here is quite remarkable. We have shown that magnons are bosonic quasiparticles and can be understood via LSWT and Fourier transforms!

1.3.2 Phase transitions and critical points

A *phase* of a physical system is understood as a state of the system which is defined by macroscopic physical properties. A *classical phase transition* is a change from one phase to another that results in the macroscopic properties of the system changing. More formally, a phase is a domain of the parameter space Γ (with parameters $g_1, g_2, \dots, g_N$) in which a function $\Phi(g_1, g_2, \dots, g_N)$ is analytic for these parameters. A phase transition is the set of points (or surfaces) at which $\Phi(g_1, g_2, \dots, g_N)$ is nonanalytic.

The 1D Ising model (cf. Section 1.2.1) does not exhibit a classical phase transition, but instead a *quantum phase transition* (QPT). A QPT can be thought of as a normal phase transition occurring at $T = 0$. There are more details to this which we shall not discuss here, but the curious reader should see [8]. Let us now restrict ourselves to consider $\Phi = e_0$, the ground state energy of the system. In this light, we can classify QPTs as we would classical phase transitions. We restrict ourselves to considering first-order or second-order QPTs in

this thesis. A first-order QPT occurs when the derivative of the ground state energy e_0 with respect to some parameter g (the parameter that is being varied) is discontinuous. That is,

$$\left. \frac{de_0}{dg} \right|_{g_c^-} \neq \left. \frac{de_0}{dg} \right|_{g_c^+}. \quad (1.16)$$

A second-order QPT is a phase transition in which de_0/dg is continuous at g_c , but a higher order derivative is discontinuous or divergent at g_c .

1.3.3 Correlation functions

An alternative method to characterizing phase transitions is by analyzing the behavior of the correlation function. The (spin-spin) *correlation function* is given as

$$C^\alpha(r) = \langle S_i^\alpha S_j^\alpha \rangle - \langle S_i^\alpha \rangle \langle S_j^\alpha \rangle, \quad (1.17)$$

where $r = |i - j|$ and $\alpha = x, y, z$. Heuristically, the correlation function provides information about how correlated sites i and j are on the spin chain. We expect the correlation function to be higher (more correlated) for small r and smaller (less correlated) for large r . For translationally invariant spin chains the term $\langle S_i^\alpha \rangle \langle S_j^\alpha \rangle = (\langle S^\alpha \rangle)^2$, which is zero if there is no net magnetization in the ground state. In this case, the correlation function becomes

$$C^\alpha(r) = \langle S_i^\alpha S_j^\alpha \rangle. \quad (1.18)$$

In the non-critical regime (e.g. not near the critical point g_c) the correlation function for gapped systems decays as

$$C^\alpha(r) \sim e^{-r/\xi}, \quad (1.19)$$

where ξ is the *correlation length* [30]. The correlation length ξ is the length scale over which spin fluctuations (or sites) are correlated. As a system approaches a second-order

QPT critical point g_c , this correlation length diverges ($\xi \rightarrow \infty$). At the critical point g_c the correlation function $C^\alpha(r)$ decays algebraically as

$$C(r) \sim r^{-\eta}, \quad (1.20)$$

where η is called the *critical exponent*. Critical exponents describe the behavior of physical quantities near the critical point and are often universal, meaning they depend only on the dimensionality of the system and the symmetries of the Hamiltonian [30].

1.4 Solitons

In our preceding discussions of one-dimensional (magnetic) systems, we have primarily focused on linear excitations that can be understood as the propagation of spin moments. These excitations are small and are typically analyzed with (linear) harmonic approximations. However, many interesting phenomena in these systems, particularly those with strong interactions, are governed by nonlinear dynamics. This section introduces a unique class of nonlinear excitations called *solitons*. Solitons are stable, localized wave packets that maintain their shape and velocity upon propagation and even after collision with other solitons. Most relevant to the content of this thesis, solitons can manifest in one-dimensional systems as *domain walls*. This section will discuss the mathematical properties of solitons (including their topological properties) and their connection to field theory. This section will be the foundation for understanding later numerical results.

1.4.1 Integrable solitons

An *integrable soliton* is a "soliton solution" to an integrable nonlinear partial differential equation. Many nonlinear partial differential equations (PDEs) in physics admit soliton solutions. For our purposes, we will use the nonlinear Schrödinger equation (NLS) as the primary example to illustrate these solutions. The nonlinear Schrödinger equation for a

spatially homogeneous system is given as

$$i\frac{\partial}{\partial t}\psi(x,t) + \frac{\partial^2}{\partial x^2}\psi(x,t) + g|\psi(x,t)|\psi(x,t) = 0, \quad (1.21)$$

where $\psi(x,t)$ is a complex field and $g > 0$. It is well known that (1.21) is *integrable* for $g > 0$ via the inverse scattering transform, and admits *soliton* solutions [31]. In fact, there is an entire class of nonlinear wave equations (because the solutions to these differential equations look like waves) that are integrable and admit soliton solutions. In the case of the nonlinear Schrödinger equation (1.21), the soliton solution is

$$\psi(x,t) = \eta \operatorname{sech}(\eta(x - ct)) \exp\left(icx - it\left(\frac{c^2}{2} - \frac{\eta^2}{2}\right)\right), \quad (1.22)$$

where c is the velocity of the wave and η is an amplitude parameter. It is important to note that the soliton solution (1.22) is an *integrable soliton*, not a *topological soliton*. It is the latter that we concern ourselves with for the forthcoming discussion.

1.4.2 Homotopy of maps, path-connectedness, and contractible spaces

To understand the concept of topological solitons, we first need to introduce some basic ideas from homotopy theory. Homotopy theory is a branch of algebraic topology concerned with classifying topological spaces based on the "holes" or "connectivity" they possess.

Let X and Y be topological spaces. Two continuous maps $f, g : X \rightarrow Y$ are said to be *homotopic*, denoted $f \simeq g$, if there exists a continuous map $H : X \times [0, 1] \rightarrow Y$ such that $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$ for all $x \in X$. The map H is called a *homotopy* between f and g . Intuitively, f can be continuously deformed into g through the family of maps $h_s(x) = H(x, s)$ for $s \in [0, 1]$. The relation $\simeq$ is an equivalence relation on the set of continuous maps from X to Y . The intuitive picture for two homotopic curves $f(x)$ and

$g(x)$ (for $x \in [0, 1]$) is seen in Figure 1.2.

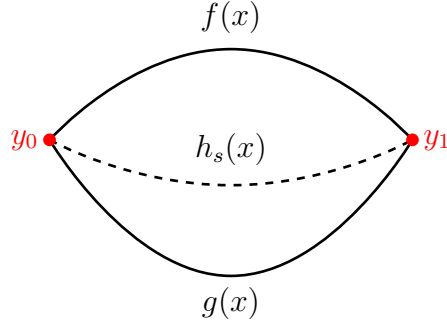


Figure 1.2: Homotopy of two curves $f(x)$ and $g(x)$ with $x \in [0, 1]$. The dotted curve $h_s(x)$ continuously interpolates between $f(x)$ and $g(x)$ for varying $s \in [0, 1]$. Note that $h_1(x) = f(x)$ and $h_0(x) = g(x)$.

A *path* in a topological space Y is a continuous map $\gamma : [0, 1] \rightarrow Y$. The points $\gamma(0)$ and $\gamma(1)$ are the start and end points of the path, respectively. A space Y is *path-connected* if for any two points $y_0, y_1 \in Y$, there exists a path γ in Y such that $\gamma(0) = y_0$ and $\gamma(1) = y_1$.

A space Y is *contractible* if its identity map $\text{id}_Y : Y \rightarrow Y$ (where $\text{id}_Y(y) = y$) is homotopic to a constant map $c_{y_0} : Y \rightarrow Y$ (where $c_{y_0}(y) = y_0$ for some fixed $y_0 \in Y$). Intuitively, a contractible space can be continuously shrunk to a single point within itself. As an example, $\mathbb{R}^n$ is contractible for any n . It is easy to see that any convex set is contractible in the topological sense. Note lastly that a point is trivially contractible. Contractible spaces have no "holes" at any dimension, and most notably, have trivial fundamental group [32].

1.4.3 Fundamental group

The fundamental group provides an idea of 1-dimensional "holes" in a topological space. Consider a topological space X and pick a base point $x_0 \in X$. A *loop* based at x_0 is a path $\alpha : [0, 1] \rightarrow X$ such that $\alpha(0) = \alpha(1) = x_0$. Two loops α and β based at x_0 are *path-homotopic* if there is a homotopy $H : [0, 1] \times [0, 1] \rightarrow X$ such that $H(s, 0) = \alpha(s)$, $H(s, 1) = \beta(s)$ for all $s \in [0, 1]$, and $H(0, t) = H(1, t) = x_0$ for all $t \in [0, 1]$. Intuitively, this means one loop can be deformed into the other while keeping the base point fixed. This

defines an equivalence relation on the set of loops based at x_0 . Let $[\alpha]$ denote the equivalence class of the loop α .

The set of these homotopy classes of loops based at x_0 is denoted $\pi_1(X, x_0)$, and can be given a group structure [32]. $\pi_1(X, x_0)$ is called the *fundamental group* of X with base point x_0 . If X is path-connected, the fundamental group is independent of the choice of base point x_0 (up to isomorphism), and we often write it as $\pi_1(X)$ [33].

Before proceeding with some examples, we provide the definition of a simply connected space. A space X is *simply connected* if it is path-connected and $\pi_1(X)$ is the trivial group. Intuitively, a simply connected space has no 1 dimensional "holes." As we shall see, a space being simply connected is a useful criterion for understanding the topology of the vacuum manifold and hence the existence of topological solitons.

Example 1.4.1. Let $S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$. A loop in S^1 can be characterized by how many times it winds around the circle, and in which direction. For example, the loop $\alpha_n : [0, 1] \rightarrow S^1$ given by $\alpha_n(s) = (\cos(2\pi ns), \sin(2\pi ns))$ winds n times counter-clockwise if $n > 0$, clockwise if $n < 0$, and is constant if $n = 0$. Two loops are homotopic if and only if they have the same winding number [33]. Thus, $\pi_1(S^1) \cong \mathbb{Z}$.

Since $\pi_1(S^1)$ is not trivial, we observe that S^1 is not simply connected.

Example 1.4.2. Let $S^2 = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$. Any loop on the surface of a sphere can be continuously shrunk to a point. Imagine a rubber band stretched around a sphere; it can always slide off or shrink to a point without breaking or leaving the surface. With this conceptual argument, we insist that the fundamental group of the 2-sphere is trivial, i.e., $\pi_1(S^2) \cong \{0\}$. The 2-sphere S^2 is simply connected.

1.4.4 Topological solitons

The concept of homotopy allows us to classify localized (stable) field configurations known as *topological solitons*. These solitons are not stable because of integrability, such as (1.22), but because of the topological properties of the space of possible ground states of the system.

Let us discuss the general theory first, and then narrow our focus to spin chains afterward. Consider a field theory where the field $\phi(x)$ takes values in some space $\mathcal{M}$, called the order parameter space, or more frequently the *vacuum manifold*. We are strictly interested in configurations where the field approaches one of its vacuum values at (spatial) infinity. For a system in d dimensions, compactification of the system to include $\pm\infty$ yields a map $\sigma : S_{\infty}^{d-1} \rightarrow \mathcal{M}$, where S_{∞}^{d-1} denotes the $(d-1)$ -sphere at infinity. If $\pi_{d-1}(\mathcal{M})$ is non-trivial, then there can exist field configurations that cannot be continuously deformed to a uniform vacuum state without creating a singularity or changing the boundary conditions at infinity. These non-trivial configurations are precisely what we call *topological solitons*, and the winding number (or topological charge) of the homotopy class characterizes the soliton.

Let us consider the Kitaev spin- $\frac{1}{2}$ chain as our system of choice for narrowing down our previous discussion. The field $\phi(x)$ takes half-integer values $\pm 1/2$ which correspond to $\uparrow$ or $\downarrow$, respectively. The vacuum value our spin chain field can approach is $\pm 1/2$, so the vacuum manifold $\mathcal{M}$ for the Kitaev spin chain consists of exactly two points: $\mathcal{M} = \{-1/2, +1/2\}$. In 1D, the boundary conditions at spatial infinity correspond to a map S_{∞}^0 to the vacuum manifold $\mathcal{M}$. The homotopy classes for these maps are classified by $\pi_0(\mathcal{M})$, and since $\mathcal{M}$ consists of two distinct points, we have $\pi_0(\mathcal{M}) \cong \mathbb{Z}_2$. A configuration in which $\phi(x) \rightarrow +1/2$ as $x \rightarrow -\infty$ and $\phi(x) \rightarrow -1/2$ as $x \rightarrow \infty$ (or vice versa) cannot be continuously be deformed to a uniform configuration without violating the boundary conditions. The spin chain configuration and field $\phi(x)$ are linked as:

$$\begin{aligned} |\cdots \uparrow\uparrow\uparrow\downarrow\downarrow\downarrow \cdots\rangle &\iff \begin{cases} \phi(x) \rightarrow -1/2 & x \rightarrow \infty \\ \phi(x) \rightarrow +1/2 & x \rightarrow -\infty \end{cases} \\ |\cdots \downarrow\downarrow\downarrow\uparrow\uparrow\uparrow \cdots\rangle &\iff \begin{cases} \phi(x) \rightarrow +1/2 & x \rightarrow \infty \\ \phi(x) \rightarrow -1/2 & x \rightarrow -\infty \end{cases} \end{aligned} \tag{1.23}$$

We observe that $\pi_0(\mathcal{M})$ is non-trivial and not path-connected so $\mathcal{M}$ is not simply-connected.

As shown in (1.23), these configurations represent *domain walls* in the spin chain. This is exactly what we take to be a topological soliton– a localized transition between regions with different vacuum values. The diagrammatic representation of this idea can be seen in Figure 1.3.

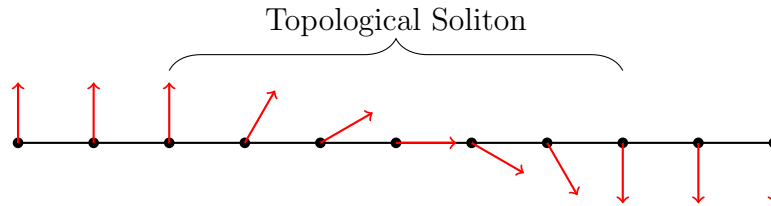


Figure 1.3: Finite spin chain showcasing a localized transition between regions of up-spin and down-spin, also known as a topological soliton.

Chapter 2

Methodology

Quantum mechanics is undoubtedly a complex theory. The many-body wavefunction of L particles $|\psi(n_1, n_2, \dots, n_L)\rangle$ has associated Hilbert space whose dimension grows on the order of d^L , where d is the local degrees of freedom of each particle. For spin- $\frac{1}{2}$ chains, the associated Hilbert space dimension grows as $\sim 2^L$ where L is the length of the spin chain. The amplitude of the wavefunction $|\psi\rangle$ that describes this system grows exponentially. Thus, one may think it is intractable to work with many-body wavefunctions in practice—but this certainly cannot be the case, as physicists work with many-body wavefunctions all the time! In condensed matter physics, the number of sites L being considered is very large. So how can physicists solve the Schrödinger equation with an exponentially complex wavefunction? In the following section we aim to answer this question and introduce Matrix Product States (MPS). Tensor networks, particularly MPS, is the formalism which modern DMRG algorithms are framed in. Throughout this section we aim to develop the theory of DMRG within the context of quantum spin chains.

2.1 Density matrix renormalization group

The density matrix renormalization group (DMRG) is a highly efficient numerical algorithm (also often called a variational method) for solving strongly correlated quantum systems in

one dimension. Although DMRG can be used for any strongly correlated quantum system, we restrict our discussion to quantum spin chains. There are two versions of DMRG: infinite-DMRG (or "iDMRG"), and finite DMRG, which we refer to hereon as just DMRG. The results of this thesis do not use iDMRG, so we provide only a brief overview of the iDMRG algorithm.

The algorithm of iDMRG is described below for an infinite spin chain with Hamiltonian $\mathcal{H}$.

Algorithm 1 iDMRG

1. Decompose the spin chain into two blocks, A and B, both with truncated Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively (see Section 2.1.1). The effective Hamiltonian for each block is $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively.
 2. Form the superblock of the chain by placing two sites between blocks A and B (call them s_1 and s_2 , respectively) and construct an effective Hamiltonian $\mathcal{H}_{\text{eff}} = \mathcal{H}_A + \mathcal{H}_{A,B} + \mathcal{H}_B$, where $\mathcal{H}_A$ is the Hamiltonian for block A, $\mathcal{H}_B$ is the Hamiltonian for block B, and $\mathcal{H}_{A,B}$ is the Hamiltonian for the interaction of sites between blocks A and B. The Hilbert space of the superblock is $\mathcal{H}_{\text{SB}} = \mathcal{H}_A \otimes \mathcal{H}_{s_1} \otimes \mathcal{H}_{s_2} \otimes \mathcal{H}_B$.
 3. Diagonalize the effective superblock Hamiltonian $\mathcal{H}_{\text{eff}}$ and find the ground state (see Section 2.1.2).
 4. Construct the reduced density operator for new blocks $A' = A \cup s_1$ and $B' = B \cup s_2$ and truncate $\mathcal{H}_A$ and $\mathcal{H}_B$ with maximal eigenvalues of the reduced density operators (see Section 2.1.3).
-

This process is shown diagrammatically in Figure 2.1.

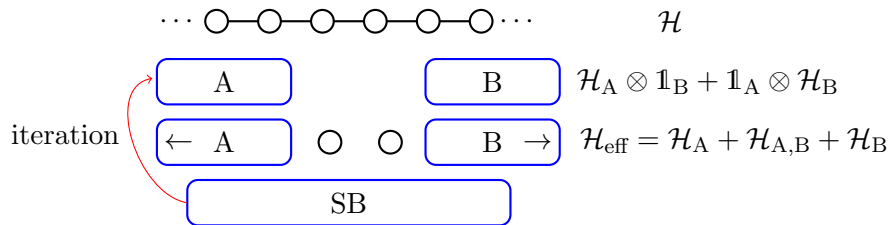


Figure 2.1: Diagram for iDMRG algorithm steps seen in Algorithm 1.

The differences between iDMRG and DMRG are subtle yet very important. In iDMRG, we introduce the two sites between blocks A and B from an infinite reservoir so that A and B

grow infinitely. However, in DMRG this is not the case. Instead, one of the blocks shrinks and the other grows to maintain a finite chain length. The algorithm of DMRG is described below.

Algorithm 2 Finite DMRG

1. Decompose the finite spin chain into two blocks, A and B of length ℓ and $L - \ell - 2$ with Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, respectively.
 2. Form the superblock of the chain as a block-site-site-block and construct an effective Hamiltonian $\mathcal{H}_{\text{eff}} = \mathcal{H}_A + \mathcal{H}_{A,B} + \mathcal{H}_B$, where $\mathcal{H}_A$ is the Hamiltonian for block A, $\mathcal{H}_B$ is the Hamiltonian for block B, and $\mathcal{H}_{A,B}$ is the Hamiltonian for the interaction of sites between blocks A and B. The Hilbert space of the superblock is $\mathcal{H}_{\text{SB}} = \mathcal{H}_A \otimes \mathcal{H}_{s_1} \otimes \mathcal{H}_{s_2} \otimes \mathcal{H}_B$.
 3. Diagonalize the effective superblock Hamiltonian $\mathcal{H}_{\text{eff}}$ and find the ground state (see Section 2.1.2).
 4. Repeat steps 1-3 by systematically by adjusting the position of the partition. That is, by incrementing the size of block B to a maximum of $L - 3$ and shrinking the size of block A to a minimum of 1, and then doing the opposite on the return sweep.
-

The DMRG algorithm is illustrated diagrammatically in Figure 2.2.

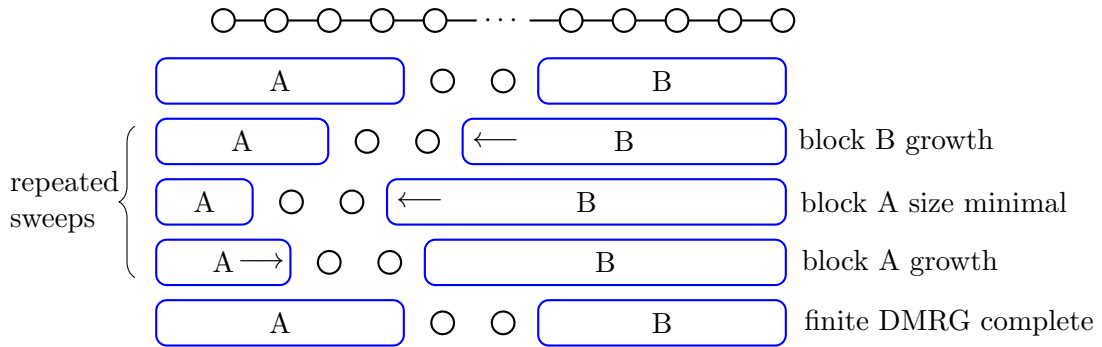


Figure 2.2: Diagram for finite DMRG algorithm steps as seen in Algorithm 2.

2.1.1 Bipartite quantum systems

Consider a spin chain of length $L \in \mathbb{Z}_{>0}$ with initial conditions left unspecified. The dimension of the associated Hilbert space $\mathcal{H}$ for this spin chain grows as $\sim d^L$, where d is the

number of degrees of freedom at each site. In the case of the spin- $\frac{1}{2}$ Kitaev spin chain, we have either spin-up or spin-down at each site, so $d = 2$ and hence the dimension scaling of the full spin chain is $\sim 2^L$.

As we shall soon see, it is very useful to partition the spin chain into left and right *blocks*. We denote these blocks A (left) and B (right). For the spin chain of length L , we can effectively partition the chain into a block-site-site-block or block-block structure (see Figure 2.3). The importance of this partition cannot be understated. The problem of addressing the exponential growth of $\mathcal{H}$ can be efficiently re-approached by truncating not the Hilbert space for the full spin chain, $\mathcal{H}$, but by truncating the associated (reduced) Hilbert spaces for block A and block B .

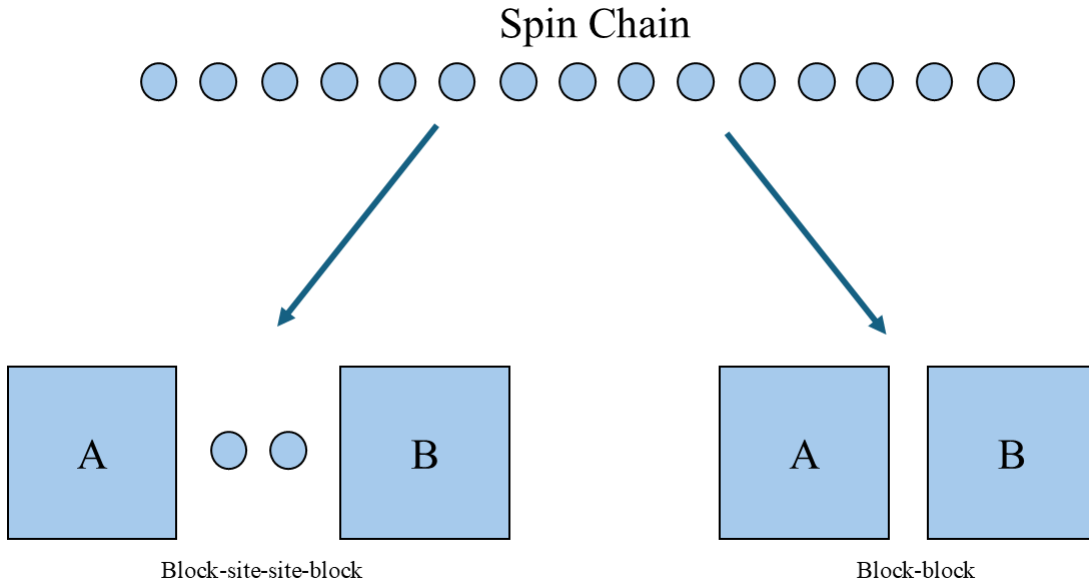


Figure 2.3: A finite spin chain can be decomposed into a block-site-site-block (left) representation or block-block (right) representation. The block-site-site-block decomposition is the only relevant decomposition for DMRG, but in principle, we can decompose any translationally invariant quantum spin chain into a bipartite system.

Suppose we have an effective description of block A of length ℓ (i.e., A has ℓ sites) in a D -dimensional (reduced) Hilbert space with orthonormal basis $|a_\ell\rangle_A$. If A has not many sites, then the basis dimension may be less than D . If A has many sites, then we assume that some truncation has occurred, and defer the discussion to later sections. It turns out

that D is a very important quantity in DMRG called the *bond dimension*. Further discussion of the bond dimension will take place in Section 2.2.

2.1.2 Ground state energy of the superblock

Suppose we are given a block-site-site-block decomposition of the Kitaev Hamiltonian constructed in Section 1.2.2, and that A and B are both of length ℓ . We call the chain of length $2\ell + 2$ the *superblock*, and now seek to determine the ground state of this superblock. Any state of the superblock ($A \bullet \bullet B$) can be written as

$$|\psi\rangle = \sum_{\langle a_A, n_A, n_B, a_B \rangle} \psi_{a_A n_A n_B a_B} |a\rangle_A |n\rangle_A |n\rangle_B |a\rangle_B \equiv \sum_{i_A j_B} \psi_{i_A j_B} |i\rangle_A |j\rangle_B, \quad (2.1)$$

where the states of the site next to A are in the set $\{|n\rangle_A\}$ of local state space dimension $d = 2$. Analogously, the states of the site next to B are in the set $\{|n\rangle_B\}$, also of local space state space dimension $d = 2$. To find the ground state energy of the superblock, we use numerical diagonalization to find the $|\psi\rangle$ that minimizes the energy

$$E = \frac{\langle \psi | \mathcal{H}_{\text{superblock}} | \psi \rangle}{\langle \psi | \psi \rangle} \quad (2.2)$$

with respect to the superblock Hamiltonian $\mathcal{H}_{\text{superblock}}$. The curious reader should see lecture notes by Noack and Manmana [34] for more details on numerical diagonalization in this context.

2.1.3 Reduced block density operator

Once the ground state of the superblock is found (see Section 2.1.2), the key step in the DMRG algorithm is to choose the most important states for representing the blocks. This choice is needed to keep the dimension of the Hilbert space manageable, as if we didn't choose the most important states then the Hilbert space dimension would grow exponentially as

discussed in Section 2. The method for optimally truncating the Hilbert space, introduced by White [21], uses the *reduced density operator*.

To define the reduced density operator, let us first consider a quantum system made of two parts, A and B. The total Hilbert space is $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. The reduced density operator for block A is given as

$$\rho_A = \text{Tr}_B |\psi\rangle \langle \psi|, \quad (2.3)$$

where the trace is taken over the degrees of freedom in block B. The reduced density operator for block B is defined similarly as

$$\rho_B = \text{Tr}_A |\psi\rangle \langle \psi|. \quad (2.4)$$

Most importantly, ρ_A acts only on $\mathcal{H}_A$ and ρ_B acts only on $\mathcal{H}_B$. In a heuristic sense, ρ_A contains all of the information that we can get from block A from the total state $|\psi\rangle$ if we do not make any measurements on block B.

The reduced density operator ρ_A is always Hermitian ($\rho_A^\dagger = \rho_A$) and positive semi-definite (meaning $\langle \phi | \rho_A | \phi \rangle \geq 0$ for any state $|\phi\rangle$ in $\mathcal{H}_A$). This means its eigenvalues, let's call them w_α , are real and not negative ($w_\alpha \geq 0$). If the total state $|\Psi\rangle$ is normalized, then the sum of these eigenvalues is given as

$$\text{Tr}(\rho_A) = \sum_{\alpha} w_{\alpha} = 1 \quad (2.5)$$

Each eigenvalue w_α tells us the probability of finding block A in the corresponding eigenstate $|u_\alpha\rangle_A$ if we were to measure block A. The reason these eigenvalues and eigenvectors are so useful in DMRG is because of how they relate to the Schmidt decomposition of $|\Psi\rangle$.

As we discuss in Appendix B, any pure state $|\Psi\rangle$ of a system with two parts A and B can be written using the Schmidt decomposition:

$$|\Psi\rangle = \sum_{\alpha=1}^r \lambda_{\alpha} |u_{\alpha}\rangle_A \otimes |v_{\alpha}\rangle_B. \quad (2.6)$$

Here, r is the Schmidt rank, which is less than or equal to the dimension of $\mathcal{H}_A$ or $\mathcal{H}_B$, whichever is smaller. The sets $\{|u_\alpha\rangle_A\}$ and $\{|v_\alpha\rangle_B\}$ are orthonormal basis states for parts of $\mathcal{H}_A$ and $\mathcal{H}_B$. The numbers λ_α are real, not negative, and are called Schmidt coefficients; they satisfy $\sum_\alpha \lambda_\alpha^2 = 1$.

If we put this Schmidt decomposition (2.6) into the definition of ρ_A from Equation (2.3), we obtain

$$\begin{aligned}\rho_A &= \text{Tr}_B \left(\sum_{\alpha,\beta} \lambda_\alpha \lambda_\beta (|u_\alpha\rangle_A \langle u_\beta|_A) \otimes (|v_\alpha\rangle_B \langle v_\beta|_B) \right) \\ &= \sum_{\alpha,\beta} \lambda_\alpha \lambda_\beta |u_\alpha\rangle_A \langle u_\beta|_A \text{Tr}_B(|v_\alpha\rangle_B \langle v_\beta|_B).\end{aligned}\tag{2.7}$$

Because the states $\{|v_\alpha\rangle_B\}$ are orthonormal, the trace over block B is $\text{Tr}_B(|v_\alpha\rangle_B \langle v_\beta|_B) = \langle v_\beta|v_\alpha\rangle_B = \delta_{\alpha\beta}$. So, the reduced density operator becomes

$$\rho_A = \sum_{\alpha=1}^r \lambda_\alpha^2 |u_\alpha\rangle_A \langle u_\alpha|_A.\tag{2.8}$$

This result (Equation (2.8)) is very important. It shows that the eigenvectors of ρ_A are exactly the Schmidt states $|u_\alpha\rangle_A$ for block A . Also, the eigenvalues w_α of ρ_A are the squares of the Schmidt coefficients, $w_\alpha = \lambda_\alpha^2$.

This connection is very useful for DMRG. When we have the reduced density operator for a large block (like $\rho_{A'}$), we can diagonalize it. Its eigenvalues $w_\alpha = \lambda_\alpha^2$ tell us how important each corresponding eigenvector $|u_\alpha\rangle_{A'}$ is. To reduce the size of the Hilbert space for A' , we choose the D eigenvectors that have the D largest eigenvalues w_α . These D states become the new, smaller, and optimized basis for representing block A' in the next DMRG steps.

2.2 Tensor Networks

2.2.1 The Illusion of Complexity

While the dimension of the Hilbert space for a spin- $\frac{1}{2}$ chain grows as 2^L , the relevant information does not span the entire Hilbert space. The way physicists can work with large ($L > 100$) systems is because of an *illusion of complexity*. The subspace containing information that we care about in the Hilbert space does not grow as 2^L , but something much more tractable. We seek to consider *physical* quantum states out of the Hilbert space. These states will have some special structure, not all of which will be discussed here. Most importantly, these so-called physical states have the property of *locality*; low energy states of quantum systems have very low entanglement. Conceptually, if we probe one point of a system, we do not expect to observe correlations far away from where we probed.

Consider a bipartite system with two parts, part A and part B . The wavefunction of this system $|\psi\rangle$ has amplitude $c_{n_1 n_2}$. The state $|\psi\rangle$ has *no entanglement* if the wavefunction amplitude can be factorized as

$$c_{n_1 n_2} = A^{n_1} A^{n_2}, \quad (2.9)$$

where A^{n_1} and A^{n_2} are single index tensors, or matrices. For an entangled state $|\psi\rangle$, $|\psi\rangle$ has *low entanglement* if the sum

$$c_{n_1 n_2} = \sum_i A_i^{n_1} A_i^{n_2} \quad (2.10)$$

has a small number of terms. The meaning of the index we sum over will be discussed in the following section.

2.2.2 Matrix product states

The whole point of tensor networks is that they provide a language to work directly with only the part of quantum mechanics—only the physical states—that have low entanglement. There are many types of tensor networks, each reflecting different entanglement geometries.

For the purposes of this thesis, we focus only on MPS, which reflect entanglement for 1D gapped systems. The idea of a MPS was first introduced by White in 1992 [21]. As we shall see later, the MPS decomposition forms the backbone of the DMRG algorithm.

A wavefunction $|\psi\rangle$ of L particles, without the use of MPS just yet, can be written as

$$|\psi\rangle = \sum_{n_1 n_2 \dots n_L} c_{n_1 n_2 \dots n_L} |n_1 n_2 \dots n_L\rangle, \quad (2.11)$$

where $c_{n_1 n_2 \dots n_L}$ is the wavefunction amplitude and $|n_1 n_2 \dots n_L\rangle$ is the set of basis states. For spin- $\frac{1}{2}$ chains, we take $|n\rangle \in \{|\uparrow\rangle, |\downarrow\rangle\}$, and for particles, we take $|n\rangle \in \{|0\rangle, |1\rangle\}$. We will call the wavefunction amplitude, which is really a set of numbers, a tensor of coefficients (of the wavefunction). Note that we do not enforce any extra conditions on $c_{n_1 n_2 \dots n_L}$ by calling it a tensor in the present moment.

Consider a lattice of L sites with d dimensional local state spaces $\{n_i\}$ on sites $i = 1, \dots, L$. From (2.11), the most general quantum state can be written as

$$|\psi\rangle = \sum_{\langle n \rangle} c_{n_1 n_2 \dots n_L} |n_1 n_2 \dots n_L\rangle, \quad (2.12)$$

where there are exponentially many coefficients $c_{n_1 n_2 \dots n_L}$. We aim to factorize the tensor of coefficients $c_{n_1 n_2 \dots n_L}$ into a product of tensors. This factorization is the MPS representation of $|\psi\rangle$, and is given by

$$|\psi\rangle = \sum_{\langle n \rangle} A^{n_1} A^{n_2} \dots A^{n_L} |n_1 n_2 \dots n_L\rangle. \quad (2.13)$$

However, from the discussion in the previous section, namely equation (2.10), we have just defined a MPS representation of $|\psi\rangle$ under the supposition that $|\psi\rangle$ has no entanglement. In practice, this is almost never the case, and systems with this property are of little to no

interest to us. Hence, a more accurate description of the MPS representation of $|\psi\rangle$ is

$$c_{n_1 n_2 \dots n_L} = \sum_{\langle a \rangle} A_{a_1}^{n_1} A_{a_1 a_2}^{n_2} A_{a_2 a_3}^{n_3} \dots A_{a_L}^{n_L}, \quad (2.14)$$

where the sum over all a_j with $j = 1, 2, \dots, L$ is called the *bond dimension* or sometimes *bond index*. Heuristically, the bond dimension provides a notion of entanglement between the sites. In the discussion of tensor networks earlier, we remarked that there are many different types of tensor networks, each capturing a different entanglement geometry. MPS is just one such tensor network that captures the entanglement for 1D (gapped) systems. The bond indices of (2.14) make this immediately obvious, as the structure of entanglement is indeed 1D and can be seen from the subscripts of the A tensors. It is also important to note that the first and last tensors ($A_{a_1}^{n_1}$ and $A_{a_L}^{n_L}$, respectively) have one fewer bond index. In the literature the bond dimension is often denoted by M , D or χ . In this thesis we denote the bond dimension as D . We make one last remark on the bond dimension: namely, that the bond dimension is at most the square root of the dimension of the associated Hilbert space. Given a Hilbert space $\mathcal{H}$ with dimension p , the bond dimension is such that $D \leq \sqrt{p}$.

2.2.3 Computing MPS

Let us begin by recalling the Singular Value Decomposition (SVD) of an arbitrary matrix M (see Appendix A):

$$M = U \Sigma V^\dagger. \quad (2.15)$$

As we shall see shortly, SVD is a staple tool for illustrating the calculations of MPS representations. The idea of using SVD to calculate the MPS representation of some function with L sites,

$$|\psi\rangle = \sum_{\langle n \rangle} c_{n_1 n_2 \dots n_L} |n_1 n_2 \dots n_L\rangle, \quad (2.16)$$

is to decompose the tensor of coefficients (with SVD) into a product of A matrices. Let us assume $|\psi\rangle$ is normalized for now. We proceed with the so-called *left-canonical MPS*. We reshape the tensor of coefficients $c_{n_1 \dots n_L}$ (which has d^L components) into a matrix Ψ of dimension $(d \times d^{L-1})$. That is,

$$\Psi_{n_1, (n_2, \dots, n_L)} = c_{n_1 \dots n_L}. \quad (2.17)$$

A SVD of Ψ yields

$$c_{n_1 \dots n_L} = \Psi_{n_1, (n_2 \dots n_L)} = \sum_{a_1}^{r_1} U_{n_1, a_1} \Sigma_{a_1, a_1} (V^\dagger)_{a_1, (n_2 \dots n_L)} \equiv \sum_{a_1}^{r_1} U_{n_1, a_1} c_{a_1 n_2 \dots n_L}. \quad (2.18)$$

The (Schmidt) rank of Ψ is $r_1 \leq d$. Our goal is to obtain the A matrices, ending up with a representation like (2.13). We decompose U into d row vectors A^{n_1} with entries $A_{a_1}^{n_1} = U_{a_1, n_1}$. Simultaneously, we reshape $c_{a_1 n_2 \dots n_L}$ into a matrix $\Psi_{(a_1 n_2), (n_3 \dots n_L)}$ of dimension $(r_1 d \times d^{L-1})$. Thus, the first step of SVD yields the (incomplete) representation

$$c_{n_1 \dots n_L} = \sum_{a_1}^{r_1} A_{a_1}^{n_1} \Psi_{(a_1 n_2), (n_3 \dots n_L)}. \quad (2.19)$$

The representation is incomplete because it does not yet look like (2.13). So, we take a SVD of Ψ again:

$$c_{n_1 \dots n_L} = \sum_{a_1}^{r_1} \sum_{a_2}^{r_2} A_{a_1}^{n_1} U_{(a_1 n_1), a_2} \Sigma_{a_2, a_2} (V^\dagger)_{a_2, (n_3 \dots n_L)} = \sum_{a_1}^{r_1} \sum_{a_2}^{r_2} A_{a_1}^{n_1} A_{a_1, a_2}^{n_2} \Psi_{(a_2 n_3), (n_4 \dots n_L)}. \quad (2.20)$$

The last equality is obtained by following the procedure for the first application of SVD. We can repeat this procedure L times, yielding a complete MPS representation:

$$c_{n_1 \dots n_L} = \sum_{a_1}^{r_1} \sum_{a_2}^{r_2} \dots \sum_{a_{L-1}}^{r_{L-1}} \sum_{a_L}^{r_L} A_{a_1}^{n_1} A_{a_1, a_2}^{n_2} \dots A_{a_{L-1}, a_L}^{n_{L-1}} A_{a_L}^{n_L} \equiv \sum_{\langle a \rangle} A_{a_1}^{n_1} A_{a_1, a_2}^{n_2} \dots A_{a_L}^{n_L}. \quad (2.21)$$

Note that assuming even L , the maximal dimensions of each A matrix in (2.13) is $(1 \times d), (d \times d^2), \dots, (d^{L/2-1} \times d^{L/2}), (d^{L/2} \times d^{L/2-1}), \dots, (d^2 \times d), (d \times 1)$, and the same applies here.

MPS are useful not only for constructing numerically efficient representations of wavefunctions, but also operators. In general, a quantum mechanical operator $\mathcal{O}$ can be written as

$$\mathcal{O} = \sum_{\substack{n_1, \dots, n_L \\ n'_1, \dots, n'_L}} c_{(n_1 \dots n_L)(n'_1 \dots n'_L)} |n_1, \dots, n_L\rangle \langle n'_1, \dots, n'_L|. \quad (2.22)$$

Thus, it is natural that we write $\mathcal{O}$ using an analogous procedure as MPS. We have

$$\mathcal{O} = \sum_{\langle n, n' \rangle} W^{n_1, n'_1} W^{n_2, n'_2} \dots W^{n_L, n'_L} |n\rangle \langle n'|, \quad (2.23)$$

where the W^{n_i, n'_i} terms are the doubly indexed analog of the A^{n_i} tensors. The details of obtaining the MPO representation will not be discussed here. The reason for this is twofold; first, the details are cumbersome and would interrupt the flow of the thesis. The second reason is because other sources do it much better (see, for example, [35] and references therein).

We should not proceed without making some remarks on the different canonical forms of MPS.

2.2.4 Canonical forms of MPS

In the previous section, we introduced the computation of the left-canonical MPS. Remarkably, there is nothing special about this choice— we can very well compute right-canonical MPS by reshaping $c_{n_1 n_2}$ into a matrix Ψ of dimension $(d^{L-1} \times d)$. That is,

$$c_{n_1 \dots n_L} = \Psi_{(n_1, \dots, n_{L-1}), n_L}. \quad (2.24)$$

The steps of SVD are exactly as before, but this time yielding right-invariant matrices $B_{a_i}^{n_i}$:

$$\begin{aligned}\Psi_{(n_1 n_2 \dots n_{L-1}), n_L} &= \sum_{a_{L-1}}^{r_1} U_{(n_1 \dots n_{L-1}), a_{L-1}} \Sigma_{a_{L-1}, a_{L-1}} (V^\dagger)_{a_{L-1}, n_L} \\ &= \sum_{a_{L-1}} \Psi_{(n_1 \dots n_{L-2}), (n_{L-1} a_{L-1})} B_{a_{L-1}}^{n_L}.\end{aligned}\tag{2.25}$$

Obtaining the full MPS representation in terms of the right-invariant matrices $B_{a_i}^{n_i}$ is exactly analogous to the left-invariant matrices $A_{a_i}^{n_i}$.

The most relevant construction to DMRG is that we can combine the left-canonical and right-canonical constructions. Suppose we did a decomposition from the left up to a site $\ell < L$ so that

$$c_{n_1 \dots n_L} = \sum_{a_\ell} (A^{n_1} \dots A_L^n)_{a_\ell} \Sigma_{a_\ell, a_\ell} (V^\dagger)_{a_\ell, (n_{\ell+1} \dots n_L)}.\tag{2.26}$$

We can reshape $V^\dagger$ as $\Psi_{(a_\ell n_{\ell+1} \dots n_{L-1}), n_L}$ and perform SVD from the right as described in the right-canonical MPS. Note that this process includes all sites up to site $\ell + 2$. Following this process through, we have

$$c_{n_1 \dots n_L} = A^{n_1} \dots A^{\ell} \Sigma B^{n_{\ell+1}} \dots B^{n_L},\tag{2.27}$$

which we call the *mixed-canonical matrix product state* of $c_{n_1 \dots n_L}$. The mixed-canonical MPS is the preferred choice for modern DMRG because it is the most computationally efficient [36].

2.3 Implementation of DMRG

In this section we briefly explain how the DMRG algorithm was implemented and performed. All code was written in Julia programming language [37]. The DMRG algorithm as described in Algorithm 2 was implemented using the ITensor Julia package [38], which handles all MPS and MPO constructions and contractions. The code was run on a personal computer.

Chapter 3

Results and Discussion

This chapter presents the numerical results obtained using the Density Matrix Renormalization Group (DMRG) method to explore the properties of the ferromagnetic and antiferromagnetic Kitaev spin chain. We analyze the ground state properties, correlation functions, and the response to external fields. Of primary interest to us is the generation of topological solitons.

3.1 Kitaev spin chain

It has recently been shown that the 1D antiferromagnetic (AFM) Kitaev spin chain with Ising interactions hosts an intermediate chiral phase with magnetic solitons [1]. For open boundary conditions, the ground state exhibits a single soliton, which lowers the energy. Our goal is to replicate the results of [1] using finite DMRG. In particular, we confirm the existence of the intermediate chiral phase hosting magnetic solitons.

3.1.1 Solitons in the spin- $\frac{1}{2}$ Kitaev chain

The soliton behavior is dependent both on the magnetic field angle as well as the magnetic field strength. For the results shown here, we restrict ourselves to the magnetic field angle(s)

$\phi_{xy} = \pi/4$ and $\theta_z = 0$. Our results agree with [1] to a limited degree. We hypothesize the difference in results is due to the computational limitations of this thesis, since DMRG was run on a personal computer.

We observe that as the magnetic field decreases, the computation time increases and accuracy decreases. Moreover, the soliton is only clearly shown for $0.55 < h < 0.72$. Beyond this, the behavior of the spin expectation values becomes unpredictable, likely due to computational limitations. This can be seen in Figure 3.1. The increased length for the $h = 0.51$

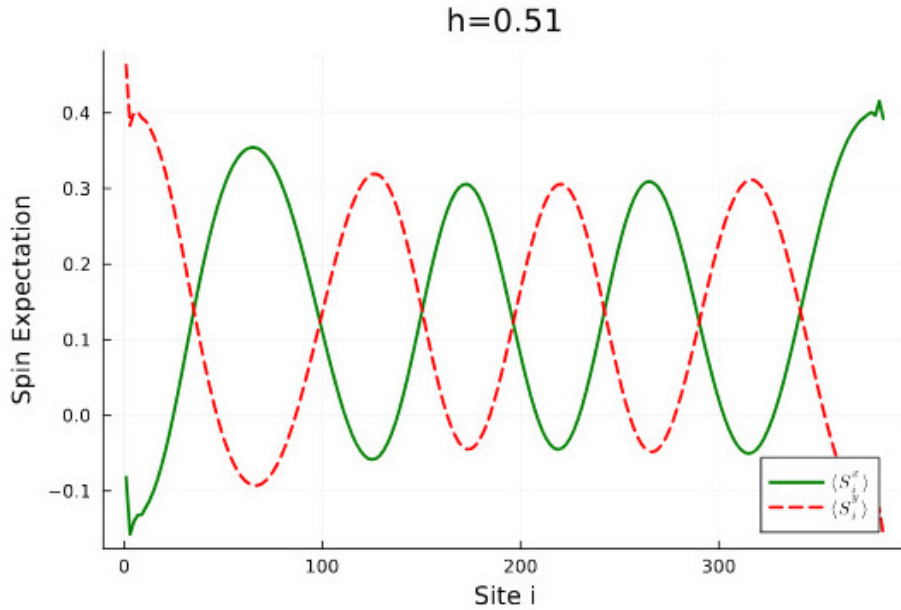


Figure 3.1: Spin expectation behavior for $h = 0.51$, $\phi_{xy} = \pi/4$, and $\theta_z = 0$ for the Kitaev chain of length $L = 384$.

case was to give the soliton, if one was present, enough space to properly form. What we observe, however, is a wildly different situation than what we expect. Instead of a soliton being completely absent, it appears that there are many solitons. If we compare the above result with the $h = 0.6$ case, as seen in Figure 3.2, we can conclude that the results of the $h = 0.51$ case should not be taken seriously until more computational power is used. When $h = 0.6$, a topological soliton is clearly formed at the midpoint of the chain. Indeed, we define a topological soliton as a localized transition from up-spins to down-spins (cf. Section 1.4.4), and the soliton in Figure 3.2 is both in accordance with our definition as well as the

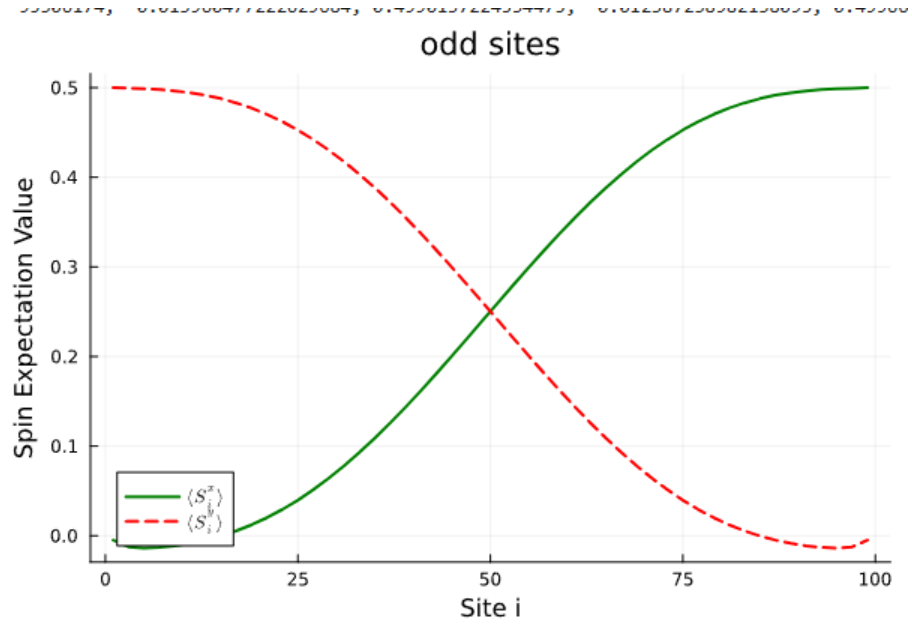


Figure 3.2: Preliminary result for $L = 99$ Kitaev chain with $h = 0.6$, $\phi_{xy} = \pi/4$, and $\theta_z = 0$ results of [1].

The $h = 0.75$ case shown in Figure 3.3 does not have a soliton, and instead, we observe the spin expectation profile to flatten out near the middle of the chain.

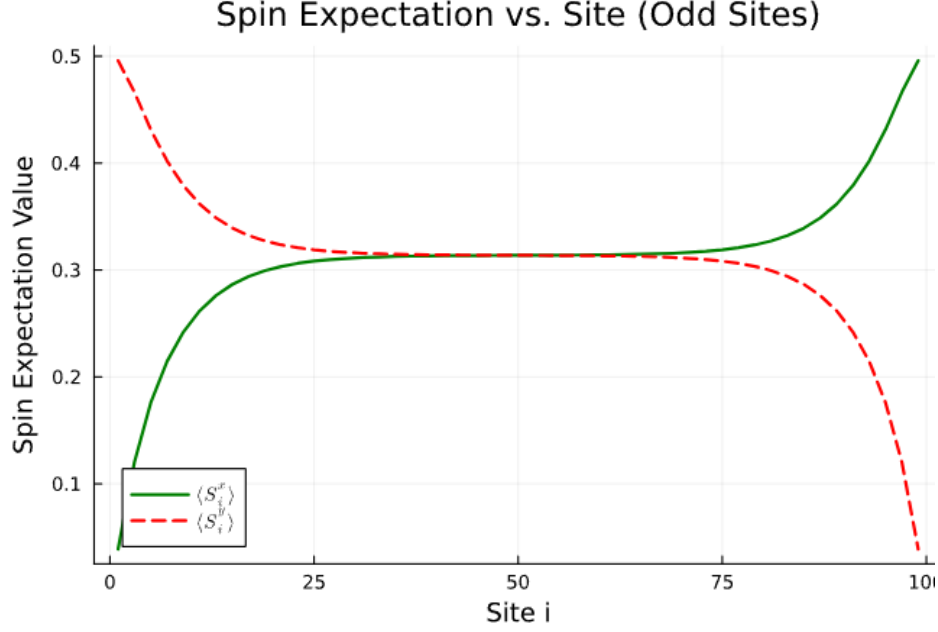


Figure 3.3: Absence of a soliton in the $L = 99$ Kitaev spin chain with $h = 0.75$, $\phi_{xy} = \pi/4$ and $\theta_z = 0$.

3.1.2 Identification of soliton phase

We confirm the existence of a soliton phase in Figure 3.4. In accordance with [1], we observe the existence of a soliton phase that is dependent on both the angle of the external magnetic field ϕ_{xy} and the strength h . The soliton phase is observed to exist only for ϕ_{xy} within 40° to 50° . Moreover, the soliton phase gets smaller for increasing magnetic field values. In Section 3.1.2 we observe that for $h = 0.75$ with $\phi_{xy} = \pi/4$ and $\theta_z = 0$ the absence of a topological soliton. Thus, we extrapolate our results in Figure 3.4 to conclude that the soliton phase disappears at $h = 0.75$.

In fact, the soliton phase(s) shown in Figure 3.4 are not confined to the xy plane, which we see in Figure 3.5. The soliton phase does not extend significantly out of the xy plane. The x -axis of Figure 3.5 is less than $|5^\circ|$. What we observe in the spin- $\frac{1}{2}$ Kitaev chain, then, is a soliton phase that looks like a tight cone centered in the xy plane.

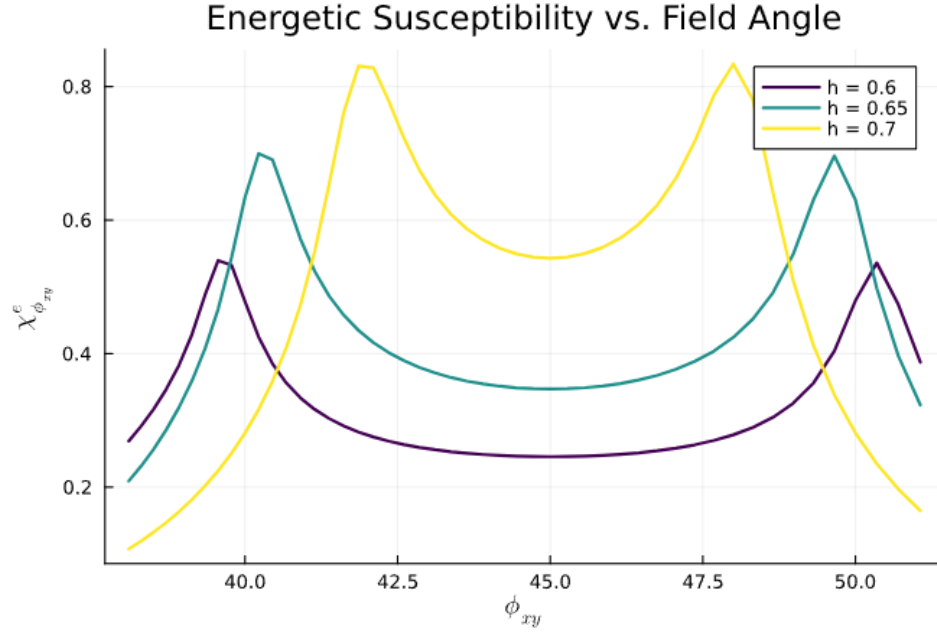


Figure 3.4: Energetic susceptibility $\chi_{\phi_{xy}}^e = -\frac{\partial^2 e_0}{\partial \phi_{xy}^2}$ is plotted vs the external magnetic field angle ϕ_{xy} in degrees with $\theta_z = 0$. The soliton phase is identified as the valley between the two peaks of each magnetic field line.

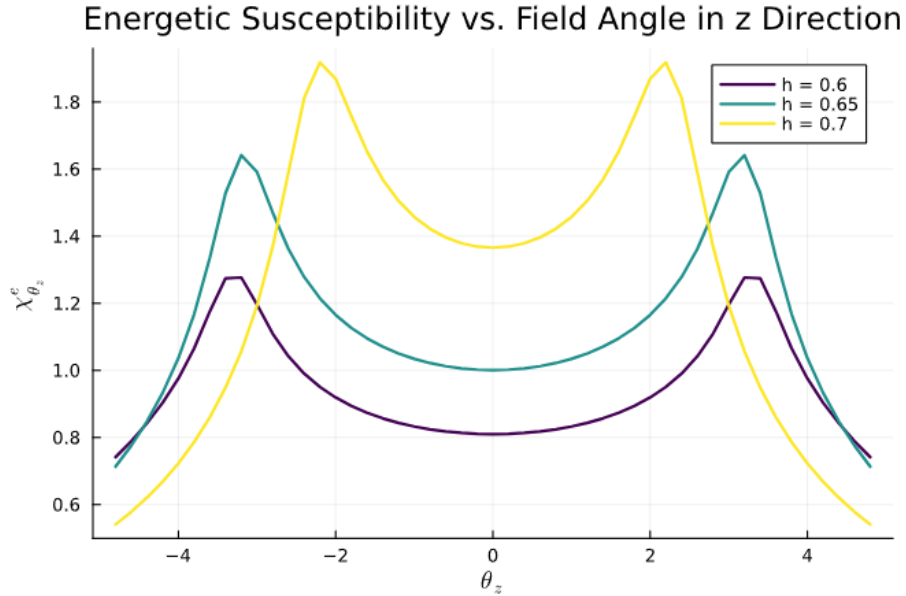


Figure 3.5: Energetic susceptibility $\chi_{\phi_{xy}}^e = -\frac{\partial^2 e_0}{\partial \phi_{xy}^2}$ is plotted vs the external magnetic field angle θ_z in degrees with $\phi_{xy} = 0$. The soliton phase is identified as the valley between the two peaks of each magnetic field line.

Chapter 4

Conclusions

This thesis set out to investigate the emergence of topological solitons in the spin- $\frac{1}{2}$ Kitaev chain using the Density Matrix Renormalization Group (DMRG) method. We focused on understanding the ground state properties under an external magnetic field and identifying the regimes where these solitonic structures appear. The numerical simulations, performed using finite DMRG as outlined in Chapter 2, have confirmed the results of existing literature [1].

As presented in Chapter 3, our DMRG calculations confirmed the formation of a topological soliton in the one-dimensional antiferromagnetic (AFM) Kitaev chain for specific magnetic field configurations. In particular, with the field angle parameters $\phi_{xy} = \pi/4$ and $\theta_z = 0$, we observed a clear soliton, characterized by a localized spin flip (cf. Section 1.4.4), within a certain range of magnetic field strengths h . A distinct soliton was visible for $h = 0.6$ (Figure 3.2). For field strengths outside this region, such as at $h = 0.75$ (Figure 3.3), the soliton was not present, and the spin expectation profile flattened at the center of the chain. At lower field strengths, for instance $h = 0.51$ (Figure 3.1), the results became less clear. We attribute this behavior primarily to the computational limitations encountered, which may have prevented the system from fully converging to the true ground state or resolving the soliton structure accurately.

It is important to acknowledge the limitations faced during this thesis. The primary constraint was the available computational power (a personal computer), which restricted the maximum bond dimension D and the system sizes L that could be simulated for ranges of parameters. This impacted the precision of the results in regimes near criticality, such as at low magnetic field strengths.

There is one primary area of future research that should be pursued. Repeating the DMRG simulations with greater computational resources would be highly beneficial. This could help clarify the spin configurations observed at lower field strengths (e.g., $h = 0.51$) and would allow for the use of larger system sizes and bond dimensions. Extending the techniques used in this thesis to investigate solitons in spin chains with ladder geometries appears to be a promising direction [personal correspondence with Axel Saenz Rodriguez]. The spin- $\frac{1}{2}$ Heisenberg-Ising XXZ chain in a ladder configuration could have novel dynamics arising from linear excitations (magnons) and nonlinear excitations (solitons).

This thesis has successfully used the DMRG technique to explore and identify topological solitons in the spin- $\frac{1}{2}$ Kitaev chain. Despite the limitations on computational resources, the results of this thesis confirm the existence of the soliton phase seen in [1].

Chapter 5

Acknowledgments

I would like to thank my advisor Dr. Axel Saenz Rodriguez for helping with this thesis and for being an overall great mentor.

Chapter 6

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Appendix A

Singular Value Decomposition

Singular Value Decomposition (SVD) is one of the most versatile tools in linear algebra. In a heuristic sense, SVD diagonalizes non-square matrices. SVD guarantees that for any arbitrary matrix M of dimension $(N_1 \times N_2)$ the existence of a decomposition

$$M = U\Sigma V^\dagger. \tag{A.1}$$

The matrix U is of dimension $(N_1 \times \max(N_1, N_2))$ and such that $U^\dagger U = \mathbb{1}$. If $N_1 \leq N_2$ then U is unitary, and so $UU^\dagger = \mathbb{1}$ as well. The matrix Σ is a rectangular diagonal matrix of dimension $(\min(N_1, N_2) \times \min(N_1, N_2))$. The diagonal entries $\sigma_i = \Sigma_{ii}$ are uniquely determined by M and called the *singular* values of M . The number of nonzero singular values, denoted by r , determines the (Schmidt) rank of M . The matrix $V^\dagger$ is of dimension $(\min(N_1, N_2) \times N_2)$ and such that $V^\dagger V = \mathbb{1}$. If $N_1 \geq N_2$ then V is unitary, and so $VV^\dagger = \mathbb{1}$ as well.

There are innumerable cases in which the properties of singular values and vectors can be utilized to our benefit. One case of practical importance is in approximating a matrix M

of rank r by a matrix M' of rank $r' < r$. We approximate M by M' in the Frobenius norm,

$$\|M\|_{\text{F}}^2 = \sum_{ij} |M_{ij}|^2, \quad (\text{A.2})$$

which is induced by the inner product $\langle M|N \rangle = \text{Tr } M^\dagger N$. Our approximation is given by

$$M' = U \Sigma' V^\dagger, \quad \Sigma' = \text{diag}(\sigma_1, \sigma_2, \dots, \sigma_{r'}, 0, \dots), \quad (\text{A.3})$$

i.e., one sets all but the first r' singular values to be 0. In the discussion of DMRG in Section 2, we use this same idea to shrink the column dimension of U and the row dimension of $V^\dagger$ to r' .

Appendix B

Schmidt Decomposition

One use for SVD is in deriving the so-called *Schmidt decomposition*. Consider a bipartite Hilbert space $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ with $\mathcal{H}_A$ of dimension N_1 and $\mathcal{H}_B$ of dimension N_2 . Suppose now that $\mathcal{H}_A$ has orthonormal basis $\{u_1, u_2, \dots, u_{N_1}\}$ and $\mathcal{H}_B$ has orthonormal basis $\{v_1, v_2, \dots, v_{N_2}\}$. Any vector $w \in \mathcal{H}$ can then be expressed as

$$w = \sum_{ij} \beta_{ij} u_i \otimes v_j. \quad (\text{B.1})$$

If we do an SVD of β_{ij} , we obtain

$$\begin{aligned} w &= \sum_{ij} \sum_{a=1}^{\min(N_1, N_2)} U_{ia} \Sigma_{aa} V_{ja}^\dagger u_i \otimes v_j \\ &= \sum_{a=1}^{\min(N_1, N_2)} \left(\sum_i U_{ia} u_i \right) \left(\sum_j V_{ja}^\dagger v_j \right) \\ &= \sum_{a=1}^{\min(N_1, N_2)} \sigma_a u_a \otimes v_a. \end{aligned} \quad (\text{B.2})$$

If we restrict the sum to run over all $r \leq \min(N_1, N_2)$ singular values, we obtain the *Schmidt decomposition* (of w),

$$w = \sum_{a=1}^r \sigma_a u_a \otimes v_a. \quad (\text{B.3})$$